



I Outline

Analisis Data

1. Dataset C-MAPSS
2. Pembersihan Data
3. Normalisasi
4. Seleksi Fitur (Pearson + PCA)

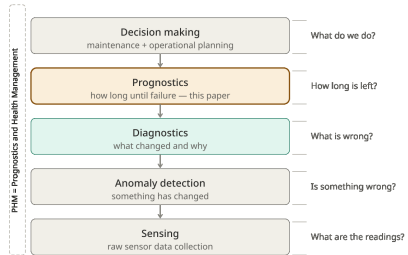
Pemodelan

- start=5 Model ANN (Python + MATLAB)
- stbrt=5 Benchmark 8 Model
- stcrt=5 Analisis Garson
- stdrt=5 Perbandingan Python vs MATLAB
- stert=5 Kesimpulan

Raw .txt → HDF5/CSV → Cleaning → Normalisasi → PCA → **ANN** → Garson

Latar Belakang

- **RUL** (*Remaining Useful Life*) = sisa siklus sebelum kegagalan komponen
- Digunakan dalam sistem **PHM** (*Prognostics and Health Management*)
- Tujuan: optimalkan jadwal perawatan, kurangi *downtime* tak terencana
- Dataset: **C-MAPSS NASA** — simulasi turbofan dari kondisi sehat hingga gagal



Pertanyaan: Dari pembacaan sensor saat ini, berapa siklus tersisa sebelum mesin gagal?

Dataset C-MAPSS

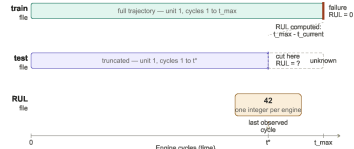
Dataset	Conditions op settings	Fault modes degraded modules	Train trajectories	Test trajectories	RUL file ground truth
FD001 easiest	1 sea level only	1 HPC degradation	100	100	RUL_FD001.txt 100 values
FD002 medium	6 alt / Mach / TRA	1 HPC degradation	260	259	RUL_FD002.txt 259 values
FD003 medium	1 sea level only	2 HPC + Fan	100	100	RUL_FD003.txt 100 values
FD004 hardest	6 alt / Mach / TRA	2 HPC + Fan	248	249	RUL_FD004.txt 249 values

Average Susceptors

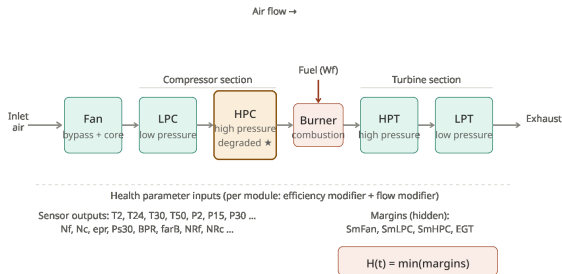
Conditions axis: FD001/FD003 (1 condition) vs FD002/FD004 (6 conditions)
 Fault mode axis: FD001/FD002 (1 fault mode) vs FD003/FD004 (2 fault modes)

Files per sub-dataset: train_FDxxx.txt test_FDxxx.txt RUL_FDxxx.txt (12 files total)
 26 columns in train/test (unit, cycle, 3 op settings, 21 sensors). RUL file: one integer per engine.

- 4 sub-dataset: FD001–FD004
- Kondisi operasi: 1 atau 6
- Mode kerusakan: HPC, atau HPC+Fan
- 26 kolom: unit, cycle, 3 op_settings, 21 sensor
- **Train:** siklus penuh hingga gagal → RUL dapat dihitung
- **Test:** dipotong sebelum gagal → RUL harus diprediksi

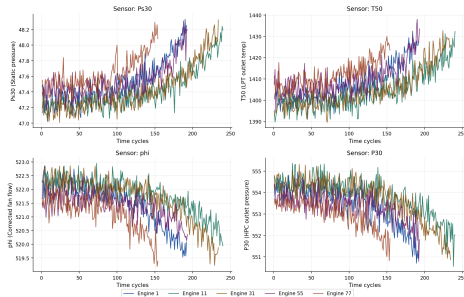


Struktur Mesin dan Sinyal Degradasi



21 sensor mengukur kondisi sepanjang aliran udara

Sensor Degradation Trajectories — C-MAPSS FD001
Each line = one engine, from first cycle to failure



Sensor *drift* menjelang kegagalan

5 Metode Analisis:

1. Standar deviasi (std)
2. Koefisien variasi (CV)
3. Korelasi Pearson vs RUL
4. Korelasi Spearman vs RUL
5. Mutual Information vs RUL

Hasil per sub-dataset:

Sub-dataset	Dihapus	Sensor Aktif
FD001	6	15

Sensor dieliminasi FD001:

T2, P2, epr, farB, Nf_dmd, PCNfR_dmd

Alasan: $\text{std} = 0 \Rightarrow \text{Pearson/Spearman} = \text{NaN}$

FD002/FD004: tidak ada eliminasi karena 6 kondisi membuat semua sensor bervariasi

Catatan P15: Pearson = $-0,13$ (hampir nol) $\Rightarrow$ dieliminasi dari FD001. Namun Garson menemukan sinyal non-linear yang tersembunyi.

Normalisasi — Z-Score Berbasis Siklus Sehat

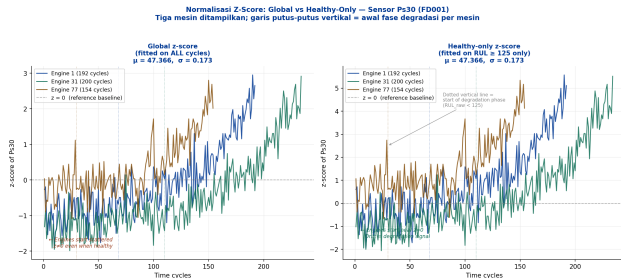
Formula:

$$z = \frac{x - \mu_{c, \text{healthy}}}{\sigma_{c, \text{healthy}}}$$

Tiga prinsip:

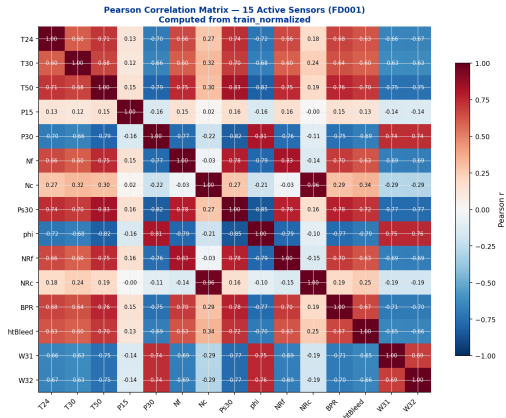
1. Parameter dari siklus *healthy* ($RUL \geq 125$) saja
2. Per kondisi operasi (bukan global)
3. Parameter *train* diterapkan ke *test* — hindari *data leakage*

Hasil: $z \approx 0$ = sehat, z menyimpang = terdegradasi

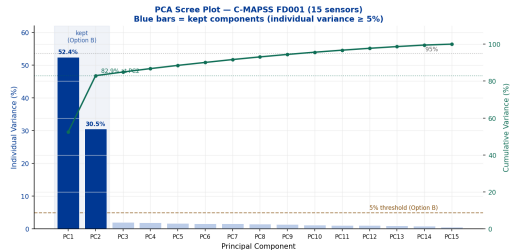


Kanan: ketiga mesin mulai di $z \approx 0$, *drift* seiring degradasi

Seleksi Fitur — Pearson & PCA

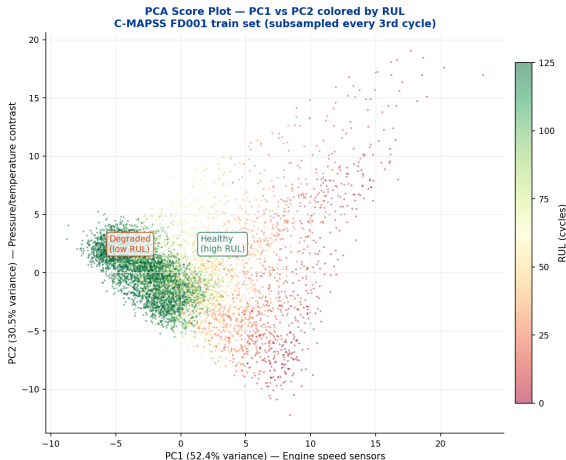


Korelasi Pearson — Nc–NRc $r = 0,96$
(redundansi tinggi $\Rightarrow$ motivasi PCA)



Scree plot — *cliff* setelah PC2; 2 komponen =
82,9% varians

PCA — Separasi Sehat vs Terdegradasi



- Setiap titik = satu siklus mesin
- **Warna** = RUL (pseudo-Z-axis)
- Hijau (sehat) → kanan
- Merah (degraded) → kiri

Dataset	Komp.	Kumulatif
FD001	2	82,9%
FD002	4	74,8%
FD003	3	87,8%
FD004	3	77,5%

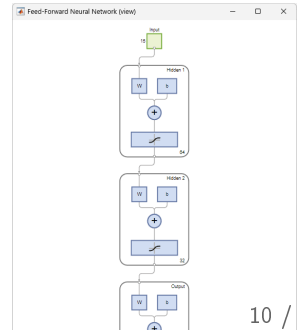
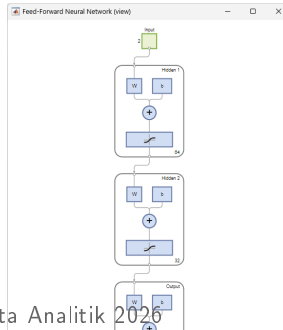
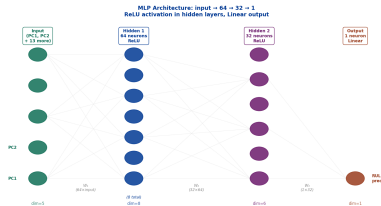
PC1 = kecepatan poros (Nf, NRf, Nc, NRc)

PC2 = kontras tekanan-kecepatan

Model ANN — Arsitektur MLP

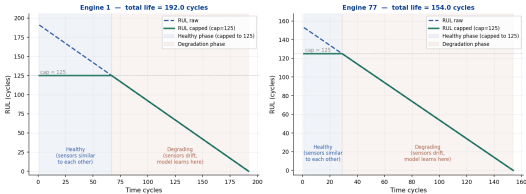
Konfigurasi

Arsitektur	input $\rightarrow 64 \rightarrow 32 \rightarrow 1$
Target	RUL_capped (cap = 125)
Epoch	500
Batch	512
Python	Adam, ReLU
MATLAB	trainscg, tansig



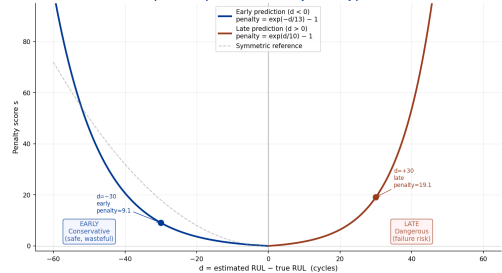
RUL Cap dan NASA Score

RUL Raw vs RUL Capped (cap = 125) – C-MAPSS FD001
Cap removes uninformative early phase; model focuses on degradation



Cap = 125: fase sehat diratakan; model fokus pada fase degradasi

NASA PHM08 Asymmetric Scoring Function
Late predictions penalised more heavily than early predictions

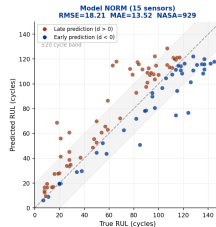
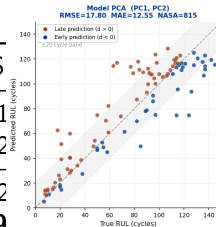


Prediksi *late* dihukum lebih berat karena membahayakan keselamatan

Hasil Benchmark — 8 Model Python

Dataset	Input	Dim	RMSE	MAE	NAS
FD001	PCA	2	17,80	12,55	81
FD001	NORM	15	18,21	13,52	92
FD002	PCA	4	29,30	20,52	14.42
FD002	NORM	21	29,45	20,73	12.99
FD003	PCA	3	21,87	16,17	2.650
FD003	NORM	16	20,16	15,28	1.78
FD004	PCA	3	30,59	23,28	9.26
FD004	NORM	21	29,68	22,38	7.15

True vs Predicted RUL — 100 Test Engines (FD001)
Each point = one engine at its last observed cycle

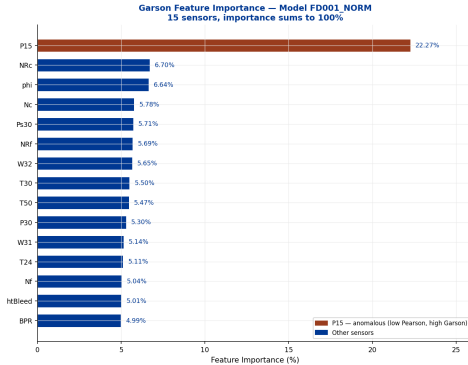


Faktor dominan: kondisi operasi — 6 kondisi
⇒ RMSE 60% lebih tinggi

PCA lebih baik: dataset sederhana

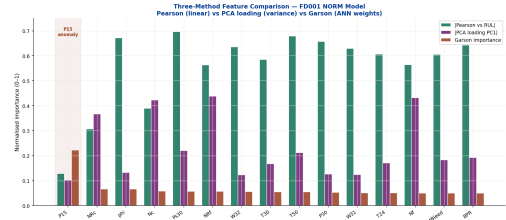
NORM lebih baik: dataset kompleks

Analisis Garson — Kepentingan Fitur



dominan (22,27%) meski Pearson = $-0,13$

P15



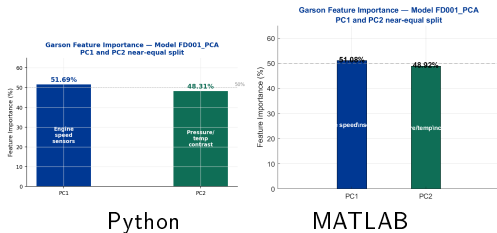
Tiga metode: Pearson rendah, PCA rendah,
Garson tinggi — ANN menemukan sinyal
non-linear

Perbandingan Python vs MATLAB

Impl.	Model	RMSE	MAE	NASA
Python	PCA	17,80	12,55	815
Python	NORM	18,21	13,52	929
MATLAB	PCA	18,30	13,34	845
MATLAB	NORM	20,91	15,75	1.319

PCA: selisih $< 3\%$ — konsisten

NORM: selisih 14,8% — ReLU vs tansig

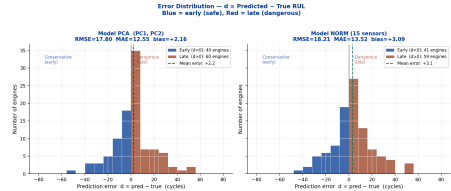


- **Optimizer:** Adam (Python) vs trainscg (MATLAB)
- **Aktivasi:** ReLU vs tansig
- **GPU:** PyTorch RTX; feedforwardnet CPU only
- **Garson PCA:** konsisten — PC1 $\approx 51\%$, PC2 $\approx 49\%$

Implikasi: untuk penggunaan nyata, Python lebih disarankan karena akurasi lebih tinggi dan GPU support. MATLAB berguna untuk cross-validation dan dokumentasi akademik.

Kesimpulan

1. **Kondisi operasi adalah faktor dominan**
— 6 kondisi $\Rightarrow$ RMSE 60% lebih tinggi dari 1 kondisi
2. **Tipe input optimal** bergantung pada kompleksitas: PCA untuk dataset sederhana, NORM untuk kompleks
3. **Python (Adam/ReLU)** lebih akurat dari MATLAB (trainscg/tansig) terutama untuk model NORM



Best result: FD001_PCA

RMSE = 17,80 cycles

MAE = 12,55 cycles

NASA Score = 815

